

End to End SQL Project for Data Analyst

-- Create Database

CREATE DATABASE OnlineBookstore;

-- Switch to the database

\c OnlineBookstore;

-- Create Tables

DROP TABLE IF EXISTS Books;

CREATE TABLE Books (
 Book_ID SERIAL PRIMARY KEY,
 Title VARCHAR(100),
 Author VARCHAR(100),
 Genre VARCHAR(50),
 Published_Year INT,
 Price NUMERIC(10, 2),
 Stock INT

);

DROP TABLE IF EXISTS customers;

CREATE TABLE Customers (
 Customer_ID SERIAL PRIMARY KEY,
 Name VARCHAR(100),
 Email VARCHAR(100),
 Phone VARCHAR(15),
 City VARCHAR(50),
 Country VARCHAR(150)

);

DROP TABLE IF EXISTS orders;

CREATE TABLE Orders (
 Order_ID SERIAL PRIMARY KEY,
 Customer_ID INT REFERENCES Customers(Customer_ID),
 Book_ID INT REFERENCES Books(Book_ID),
 Order_Date DATE,
 Quantity INT,
 Total_Amount NUMERIC(10, 2)

);

SELECT * FROM Books;

SELECT * FROM Customers;

SELECT * FROM Orders;

-- Import Data into Books Table

```
COPY Books(Book_ID, Title, Author, Genre, Published_Year, Price, Stock)
FROM 'D:\Course Updates\30 Day Series\SQL\CSV\Books.csv'
CSV HEADER;
```

```
-- Import Data into Customers Table
COPY Customers(Customer_ID, Name, Email, Phone, City, Country)
FROM 'D:\Course Updates\30 Day Series\SQL\CSV\Customers.csv'
CSV HEADER;
```

```
-- Import Data into Orders Table
COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
FROM 'D:\Course Updates\30 Day Series\SQL\CSV\Orders.csv'
CSV HEADER;
```

-- 1) Retrieve all books in the "Fiction" genre:

```
Select * from Books where Genre = 'Fiction';
```

-- 2) Find books published after the year 1950:

```
select * from Books where Published_year > 1950;
```

-- 3) List all customers from the Canada:

```
select * from Customers where Country = 'Canada';
```

-- 4) Show orders placed in November 2023:

```
select * from Orders where Order_date between '2023-11-01' and '2023-11-30' order by
order_date;
```

-- 5) Retrieve the total stock of books available:

```
select sum(stock) from Books;
```

-- 6) Find the details of the most expensive book:

```
select * from Books order by Price DESC limit 1;
```

-- 7) Show all customers who ordered more than 1 quantity of a book:

```
select * from Orders where quantity > 1 order by quantity desc;
```

-- 8) Retrieve all orders where the total amount exceeds \$20:

```
select * from Orders where Total_Amount > 20 order by total_amount desc;
```

-- 9) List all genres available in the Books table:

```
select distinct(genre) from Books;
```

-- 10) Find the book with the lowest stock:

```
select * from Books order by stock asc limit 1;
```

-- 11) Calculate the total revenue generated from all orders:

```
select sum(Total_Amount) from Orders;
```

-- Advance Questions :

-- 1) Retrieve the total number of books sold for each genre:

```
select Distinct genre, sum(quantity)
from Books
left join Orders on Books.Book_id = Orders.Book_id;
```

-- 2) Find the average price of books in the "Fantasy" genre:

```
select round(Avg(price),2) from books where genre = 'Fantasy';
```

-- 3) List customers who have placed at least 2 orders:

```
select c.name, o.customer_id, count(o.order_id)
from Orders o
join Customers c on o.customer_id = c.customer_id
group by o.customer_id, c.name
having count(o.order_id)>1
order by count(o.order_id) desc;
```

-- 4) Find the most frequently ordered book:

```
select b.title, count(o.book_id)
from orders o
join books b on o.book_id = b.book_id
group by b.title
order by count(o.book_id) desc;
```

-- 5) Show the top 3 most expensive books of 'Fantasy' Genre :

```
select * from books
where genre ='Fantasy'
order by price desc
limit 3
```

-- 6) Retrieve the total quantity of books sold by each author:

```
select b.author, sum(o.quantity)
from orders o
join books b on o.book_id = b.book_id
group by b.author
order by sum(o.quantity) desc;
```

-- 7) List the cities where customers who spent over \$30 are located:

```
select c.city, c.country, sum(o.total_amount)
from customers c
join orders o on c.customer_id = o.customer_id
group by c.city, c.country
having sum(total_amount)>30
order by sum(total_amount)desc;
```

-- 8) Find the customer who spent the most on orders:

```
select c.customer_id, c.name, sum(o.total_amount)
from customers c
join orders o on c.customer_id = o.customer_id
group by c.customer_id,c.name
--having sum(o.total_amount)
order by sum(o.total_amount) desc
limit 5;
```

--9) Calculate the stock remaining after fulfilling all orders:

```
select b.book_id, b.title, b.stock, coalesce(sum(o.quantity),0) as order_quantity,  
b.stock - coalesce(sum(o.quantity),0) as remaining_quantity  
from books b  
left join orders o on b.book_id = o.book_id  
group by b.book_id, b.title  
order by b.book_id;
```